

An Internship Report
Of
Ethical Hacking

Submitted

To

School of Computer Science and Engineering

IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B-Tech (CSE)



By

Roll Number of Student:- 2410030397

Name of Student: SANYOG RAGHAV

Batch: 2026-27

1. CANDIDATE'S DECLARATION

I, SANYOG RAGHAV, do hereby declare that the internship report titled “**Ethical Hacking**” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature:-

Student Name: SANYOG RAGHAV

Roll No.: 24100030397

Section: 3CSE14

2. ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism, and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to thank the AICTE – EduSkills Virtual Internship Program team and EduSkills Academy for providing me the opportunity and the structured platform to learn and gain hands-on exposure in the domain of Ethical Hacking. I am also thankful to my institute, IILM University, Greater Noida, and the faculty of the School of Computer Science and Engineering for their continuous support, mentorship, and encouragement.

I express my gratitude to my parents for their blessings and constant support.

Signature:

Student Name: SANYOG RAGHAV

Roll No.: 24100030397

Section: 3CSE14

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3. INTERNSHIP COMPLETION CERTIFICATE



Internship Offer Letter



Ref No: C17Y2026M081462737

Internship Offer Letter

Student Name : Sanyog Raghav
Stream : B.Tech
Branch : B.Tech(Computer Science and Engineering)
Institute Name : IILM University, Greater Noida
Internship Domain : Ethical Hacking (english Language)
Internship Duration : June 2026 - August 2026
AICTE Student ID : STU6a2d013c18feb1781334332

We are pleased to inform you that you have been selected for the 8-Week AICTE – EduSkills Virtual Internship Program.

During this internship, you will learn and demonstrate industry-relevant skills that will enhance your employability and strengthen your confidence in the subject area.

The program is designed to provide structured learning combined with practical exposure.

Internship Structure & Evaluation:

- The internship will follow a week-wise structured learning plan.
- You will be required to complete weekly assessments to track your progress and understanding.
- Certain modules will require submission of project documentation and assignments.
- In the final week, you must appear for a Final Assessment Test.
- The final grade will be based on your overall performance, including weekly assessments, project submissions, and the final test.

Upon successful completion of all required steps and assessments, you will be awarded a Virtual Internship Certificate.

Please note that this offer letter will be considered valid only after the successful completion of the internship program and fulfillment of all assessment requirements, including obtaining the final completion certificate. Without the final completion certificate, this offer letter shall not be treated as a valid document for any official purpose.

The week wise Structure of the Internship is as Follows:

Week	Module	Description
Week 1	Introduction to Ethical Hacking	Learn the purpose, legality, tools, and techniques of ethical hacking to prevent cyber threats.
Week 2	Fundamentals of Security	Understand cybersecurity principles, threat types, and protection practices for systems and data.
Week 3	Social Engineering	Identify and prevent social engineering attacks like phishing and impersonation.
Week 4	Malware and Their Types	Learn different malware types and how to identify, analyze, and mitigate their effects.
Week 5	Networking Essentials	Understand networks, protocols, IP addressing, and basic device configurations.
Week 6	Introduction to Kali Linux	Learn Kali Linux basics, navigation, and tools for ethical hacking and penetration testing.
Week 7	Basic Web Application with Capstone Projects (Vulnerability Assessment)	Build simple web apps and assess vulnerabilities through practical projects.
Week 8	Penetration Testing (Using Kali Linux)	Perform hands-on penetration testing using Kali Linux tools and report vulnerabilities.
Capstone Project /Internship Grade Point Assessment		

Note: There is no stipend associated with this internship.

We congratulate you on your selection and wish you a successful learning journey.



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4. PROJECT DESCRIPTION

4.1 Introduction

This report presents a comprehensive account of the 8-week Virtual Internship undertaken under the AICTE – EduSkills Virtual Internship Program in the domain of “Ethical Hacking (english Language)”. The internship was carried out from June 2026 to August 2026 as part of the academic curriculum requirement for the degree of Bachelor of Technology in Computer Science and Engineering (B-Tech CSE) at IILM University, Greater Noida.

The primary objective of this internship was to bridge the critical divide between theoretical cybersecurity concepts and real-world practical offensive and defensive security operations. Throughout the 8-week structured program, the learner was immersed in industry-standard security methodologies, tools, and best practices. The training covered the complete lifecycle of ethical hacking and penetration testing—encompassing passive and active information gathering, network scanning, vulnerability identification and assessment, web application security (OWASP Top 10), system exploitation, password auditing, privilege escalation, cryptography, and incident remediation strategies.

The program was delivered through interactive modules, virtual lab environments, hands-on Capture The Flag (CTF) challenges, weekly evaluations, and an end-to-end Capstone Project. Upon satisfying all evaluation criteria, including the final comprehensive assessment, the student was awarded a Grade 'E' (Excellent) Virtual Internship Certificate by EduSkills Academy in collaboration with AICTE.

4.2 Organization Profile

EduSkills Academy is an ISO 9001:2015 and ISO 20000-1:2018 certified organization driven by the mission “Nation Building Through Skills”. Operating in partnership with the All India Council for Technical Education (AICTE), Ministry of Education, Government of India, EduSkills facilitates high-impact virtual internships under the patronage of the AICTE National Internship Portal.

EduSkills collaborates closely with global technology leaders (including Cisco, AWS, Palo Alto Networks, Fortinet, Microchip, Red Hat, and Juniper Networks) and premier academic institutions across India. The initiative designs industry-aligned curricula that equip engineering and technical students with hands-on skills in emerging technologies such as Cybersecurity, Ethical Hacking, Cloud Infrastructure, Artificial Intelligence, and DevOps, thereby closing the industry-academia readiness gap.

4.3 Problem Statement

In modern computing environments, organizations face an unprecedented frequency and sophistication of cyber threats, including zero-day exploits, ransomware campaigns, advanced persistent threats (APTs), and widespread web application vulnerabilities. While standard academic curricula typically focus on theoretical cryptographic algorithms and basic network architectures, engineering students often lack

practical exposure to how attacks actually materialize in production environments and how ethical security professionals systematically uncover and remediate flaws.

Building a secure application or network infrastructure requires thinking like an adversary to proactively identify weak points before malicious actors can exploit them. The problem addressed in this internship is the practical execution of the complete ethical hacking workflow: performing reconnaissance, mapping network attack surfaces, discovering and validating vulnerabilities in web and system services, simulating controlled exploits safely, and designing robust countermeasures and defensive hardening strategies to protect organizational assets.

4.4 Project Objectives

- To master the core principles of Cybersecurity, the CIA Triad (Confidentiality, Integrity, Availability), and ethical legal compliance frameworks.
- To perform passive and active reconnaissance (OSINT, footprinting, WHOIS, DNS enumeration) to assess target attack surfaces.
- To execute advanced network discovery, port scanning, service detection, and OS fingerprinting using Nmap and Wireshark.
- To conduct automated and manual vulnerability assessments using tools like OpenVAS, Nessus, and Nikto against network targets.
- To analyze and exploit web application vulnerabilities based on the OWASP Top 10 framework (SQL Injection, XSS, CSRF, IDOR, Broken Authentication).
- To utilize proxy inspection tools such as Burp Suite for HTTP request/response tampering and API vulnerability testing.
- To understand system hacking mechanisms, password auditing techniques (John the Ripper, Hydra), and privilege escalation pathways.
- To evaluate malware mechanisms, trojan vectors, evasion fundamentals, and modern defensive countermeasures.
- To implement defensive hardening, firewall rules, intrusion detection principles (Snort), and cryptographic protection (AES, RSA, SHA-256).
- To synthesize learned concepts into a complete end-to-end security audit and penetration testing Capstone Project.

4.5 Scope of the Project

The scope of this internship encompasses the entire ethical hacking and penetration testing methodology. This begins with pre-engagement scoping and information gathering, progresses through network/port enumeration, vulnerability scanning, web application attack surface analysis, controlled exploitation in isolated virtual environments, and concludes with post-exploitation assessment and actionable remediation reporting.

The practical exercises and testing were conducted exclusively within controlled, isolated virtual lab environments (utilizing Kali Linux, Metasploitable targets, and sandbox web applications) provided by

EduSkills Academy. All security testing strictly adhered to ethical guidelines and legal frameworks, with no unauthorized attacks conducted on live external production systems.

4.6 Technologies and Tools Used

- **Operating Systems & Lab Platforms:** Kali Linux, Parrot Security OS, Metasploitable 2/3, Ubuntu Server, VirtualBox / VMware Workstation.
- **Information Gathering & OSINT:** Whois, theHarvester, NSlookup, Dig, Shodan, Maltego.
- **Network Scanning & Packet Analysis:** Nmap (Network Mapper), Zenmap, Wireshark, TCPdump, Netcat.
- **Vulnerability Assessment:** OpenVAS, Nessus Community Edition, Nikto Web Scanner, Lynis.
- **Web Application Security:** Burp Suite Community Edition, OWASP ZAP, SQLmap, Postman, Dirb/Gobuster.
- **Exploitation & Penetration Testing:** Metasploit Framework (MSFConsole), Searchsploit, Armitage.
- **Password Auditing & Cracking:** John the Ripper, THC-Hydra, Hashcat, Crunch wordlist generator.
- **Cryptography & Defense:** OpenSSL, GnuPG (GPG), HashCalc, Snort IDS, iptables firewall.

4.7 System Architecture & Assessment Framework

The structured methodology followed during the internship corresponds to standard penetration testing execution standards (PTES) and cybersecurity defense-in-depth principles, divided into the following hierarchical stages:

- **Reconnaissance Layer:** Information gathering through passive OSINT and active DNS/domain footprinting to outline the target perimeter.
- **Scanning & Enumeration Layer:** Deep network and service scanning using Nmap to identify active hosts, open ports, protocol banners, and OS signatures.
- **Vulnerability Assessment Layer:** Automated scanning (Nessus/Nikto) and manual CVE matching to evaluate known weaknesses, misconfigurations, and outdated services.
- **Web Application Security Layer:** Inspection of HTTP headers, parameters, and form handlers using Burp Suite to audit against OWASP Top 10 vulnerabilities (SQLi, XSS, CSRF, IDOR).
- **Exploitation & System Hacking Layer:** Controlled payload crafting and execution using Metasploit Framework and proof-of-concept exploits to validate true vulnerability severity.
- **Post-Exploitation & Privilege Escalation Layer:** Assessing local privilege escalation vectors (misconfigured permissions, weak sudo rules) and data access controls.
- **Remediation & Defense Hardening Layer:** Formulating defensive patch management, firewall/IDS rules (Snort), secure coding practices, and executive vulnerability mitigation reports.

4.8 Methodology

The 8-week internship followed an intensive modular structure combining theoretical lectures, virtual lab simulations, weekly hands-on assessments, and documented project tasks, detailed in the table below:

Week	Module	Detailed Description & Practical Labs
1	Ethical Hacking Fundamentals & Lab Setup	Introduction to cybersecurity concepts, CIA Triad, hacker classifications, ethical guidelines, legal frameworks (IT Act), and configuring isolated virtual lab environments in Kali Linux and VirtualBox.
2	Reconnaissance, Footprinting & OSINT	Passive information gathering using search engines, WHOIS databases, DNS interrogation (dig, nslookup), email harvesting (theHarvester), and attack surface mapping.
3	Network Scanning & Protocol Analysis	Active host discovery, port scanning, TCP/UDP flag analysis with Nmap; live packet sniffing, protocol inspection, ARP spoofing detection, and MITM analysis using Wireshark.
4	Vulnerability Assessment & Scanning	Principles of vulnerability management, running automated scans with OpenVAS/Nessus, analyzing CVE databases, assessing Common Vulnerability Scoring System (CVSS) metrics.
5	Web Application Security & OWASP Top 10	Investigating web architecture flaws, auditing OWASP Top 10 vulnerabilities (SQL Injection, XSS, CSRF, IDOR), HTTP proxy intercepting and repeater analysis using Burp Suite and OWASP ZAP.
6	System Hacking & Privilege Escalation	System penetration techniques, exploiting known services using Metasploit Framework, password cracking with John the Ripper/Hydra, Linux/Windows local privilege escalation.
7	Malware, Cryptography & Wireless Security	Understanding malware types (Trojans, ransomware, rootkits), symmetric/asymmetric encryption, hashing algorithms (SHA, MD5), and Wi-Fi security mechanisms (WEP, WPA/WPA2 handshakes).
8	Applied Security Projects &	Full-scale penetration testing simulation on vulnerable host instances, end-to-end vulnerability assessment report writing,

Each week concluded with a weekly assessment to evaluate conceptual and technical understanding. In the 8th week, a comprehensive Final Assessment Test and practical Capstone submission were conducted, yielding an overall Grade 'E' (Excellent: 80–89%).

4.9 Expected Outcomes

- Ability to independently conduct ethical vulnerability assessments and security audits on network infrastructure.
- Hands-on proficiency with industry-grade cybersecurity tools including Kali Linux, Nmap, Wireshark, Burp Suite, and Metasploit.
- Comprehensive understanding of web application vulnerabilities (OWASP Top 10) and practical remediation techniques.
- Skill to detect, analyze, and troubleshoot network anomalies, MITM attacks, and unauthorized access attempts.
- Practical understanding of password auditing, authentication security, and defensive system hardening.
- Deep knowledge of cryptographic fundamentals (encryption, hashing, digital certificates) and their application in secure systems.
- Completion of an end-to-end penetration testing capstone project with professional security reporting and actionable mitigation roadmaps.

4.10 Certificates of Completion and Communication Proof

The Internship Completion Certificate issued by EduSkills Academy and AICTE, confirming successful completion of the 8-week AICTE – EduSkills Virtual Internship Program in Ethical Hacking (english Language) with Grade 'E' (Excellent), along with the official email communication proof, are documented



Sanyog Raghav <sanyog.raghav.cs28@iilm.edu>

AICTE-EduSkills Internship Certificate download link

1 message

EduSkills Foundation <noreply@eduskillsfoundation.org>
To: Sanyog Raghav <sanyog.raghav.cs28@iilm.edu>

Wed, Jul 29, 2026 at 7:55 PM

Dear **Sanyog Raghav**,

Congratulations!

You have successfully completed **10 weeks Ethical Hacking (english Language)**. Your certificate is ready — please log in to your profile and download it.

Steps to download your certificate:

1. Login at: <https://eduskillsfoundation.org/login>
2. Go to the **Certificate Section** from your sidebar
3. Click **Download Certificate**

in

this report.

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